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B.Tech. Degree IV Semester Regular/Supplementary Examination June 2024

CS 19-202-0402 MICROPROCESSORS
(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcomes

On successful completion of the course, the students will be able to:

- CO1: Describe the architecture and organization of 8085 and 8086 Microprocessor.
 CO2: Understand and classify the instruction set of 8085/8086 microprocessor and distinguish the use of different instructions and apply it in assembly language programming.
 CO3: Relate the addressing modes used in the instructions.
 CO4: Realize the Interfacing of memory and various I/O devices with 8085/8086 microprocessor.
 CO5: Familiarise the architecture and operation of Programmable Interface Devices and realize the programming and interfacing of it with 8085 microprocessor.
 CO6: Interface various peripheral IC's with Intel 8085/8086 microprocessor for its various applications.
 Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create
 PO – Programme Outcome

PART A(Answer **ALL** questions)

	(8 × 3 = 24)	Marks	BL	CO	PO
I. (a)	How does the 8085 microprocessor differentiate between address and data on the multiplexed bus?	3	L2	2	1,2
(b)	How do the INTA and RST signals function in the 8085 microprocessor?	3	L2	1	1,2
(c)	Explain the difference between JMP and CALL instructions in the 8085.	3	L1	1	1
(d)	How do wait states affect the instruction timing in the 8085?	3	L2	3	1,2,4
(e)	Explain flag register in 8086.	3	L2	5	2
(f)	What are assembler directives and how are they different from instructions in the 8086?	3	L2	5	1,3
(g)	What is the difference between Mode 0, Mode 1 and Mode 2 operations of 8255?	3	L1	8	1,3
(h)	What challenges are involved in interfacing high-speed memory with the 8086 and how can they be addressed?	3	L2	8	1,3

PART B

(4 × 12 = 48)

II. (a) With the help of block diagram, explain the internal architecture of 8085 microprocessor.	7	L2	2	1
(b) How does the 8085 handle multiplexing and demultiplexing of the address/data bus?	5	L1	1	1

OR

III. (a) Draw the pin diagram and explain functional details of pins in 8085 microprocessor.	7	L1	1	1
(b) Explain the bus structure of the 8085.	5	L2	2	1

(P.T.O.)

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		Marks	BL	CO	PO
IV.	(a) Discuss the key differences between memory-mapped I/O and I/O-mapped I/O in the 8085 microprocessor.	6	L3	4	1,3,4
	(b) Write an 8085 assembly language program to find the largest number in an array of 10, 8-bit numbers stored from memory location 4000H to 4009H. Store the largest number in memory location 4010H.	6	L1	4	1,3,4
OR					
V.	(a) Explain the concept, types, prioritization and handling of interrupts in 8085.	6	L3	4	1,3,4
	(b) Draw and explain the timing diagram for STA 2000H.	6	L1	4	1,3,4
VI.	(a) With the help of pin diagram, explain in detail about the maximum mode of operation in 8086 and draw memory read timing diagram in maximum mode.	8	L2	6	1,3,4
	(b) Explain addressing modes of 8086 and give examples.	4	L2	7	1,3,4
OR					
VII.	(a) Draw the pin diagram and explain functional details of pins in 8086 microprocessor.	7	L2	6	1,3,4
	(b) Write an 8086 assembly language program to calculate the factorial of a number stored in memory location 2050H. Store the result in memory location 2052H.	5	L2	7	1,3,4
VIII	(a) Explain:	8	L1	8	1,3
	(i) The modes of operation supported by 8253 PIT				
	(ii) The control and status registers used in the 8259 PIC interfacing with the 8086.				
	(b) Write an 8085 assembly language program to initialize the 8279 keyboard/display controller in a simple display mode, read a key from the keyboard and display it on a connected display unit. Assume the 8279 is mapped to memory address 6000H.	4	L2	8	1,3
OR					
IX.	(a) Explain the initialization sequence required to configure the DMA Controller 8257 for interfacing with the 8085, its data transfer process between 8085 and advantages of using 8257 in conjunction with 8085.	8	L2	8	1,3
	(b) Describe the concept, operation of BSR mode in the 8255 PPI, advantages and its applications.	4	L1	8	1,3

Bloom's Taxonomy Levels

L1 - 33.3%, L2 - 58.33%, L3 - 8.3%.
